

Biologists Discover How Vitamin C Helps Fish

Scientists have known for years that fish use vitamin C for proper bone growth and to increase their tolerance to environmental stresses. Yet, they never knew exactly how this occurred. Now they do, and the discovery may help biologists better combat the effects of pollution on fish, the Department of the Interior announced today.

Biologists at the U.S. Fish and Wildlife Service's Fish Pesticide Laboratory in Columbia, Missouri, first learned of the mechanism while studying channel catfish that were affected by the insecticide toxaphene. This pesticide is widely used on cotton crops in the South where channel catfish are also raised commercially.

About 6 years ago, biologists noticed that the channel catfish grown on fish farms in that area were developing a curvature of the spine that, in extreme cases, broke the fish's back and stunted growth as much as one-third. Last year, Fish and Wildlife Service scientists linked this syndrome to toxaphene residues in the water. Concentrations as low as 37 parts per trillion in the water were found to have serious long-term effects on catfish. Earlier this year, they documented for the first time the specific role vitamin C plays in this process, and the implications are significant for future fish culture as well as for a better understanding of the chronic effects of pollutants on fish.

The mechanism works this way: Vitamin C is used by fish in a number of ways and in various parts of the body. One primary function of the vitamin is to aid in the formation of collagen—the protein framework or base upon which bone develops. Calcium and phosphate minerals are deposited within and around this framework forming a skeleton. An insufficient supply of Vitamin C can increase the mineral ratio making the backbone brittle and finally snapping it, which can result in internal bleeding. If the fish survives, its growth is severely stunted. In addition, with a decrease of collagen, the ability of fish to heal wounds or regenerate tissue is affected. Finally, vitamin C is an essential nutrient of the liver and is used as a key defense to detoxify poisonous substances in the environment. Without it, fish cannot respond as well nor adapt to stresses.

Biologists at the Service's Fish Pesticide Research Laboratory found that when catfish were chronically exposed to toxaphene residues, even in trace amounts, their ability to form collagen was inhibited. Research showed that most of the vitamin C is diverted to the liver where it is used to neutralize the effect of toxaphene. So much of it is diverted that there isn't enough of the nutrient left for other metabolic processes.

The result is a functional

deficiency of the vitamin, even though the amount may be the same as in a healthy fish, and reduced growth and the broken back condition can occur. Brook trout and fathead minnows exhibit similar symptoms when exposed to low concentrations of toxaphene.

Although there is no known method to increase the vitamin C content of fish in the wild, the discovery of its importance in fish metabolism has been quite beneficial to the multi-million dollar fish farming business of the South where up to 5 percent of pond-cultured catfish showed the symptom. Vitamin C is now included in many commercially prepared fish foods and its inclusion in the diet of farmed catfish is helping to eliminate the "broken back syndrome."

How ever, the discovery of vitamin C's importance in fish metabolism is equally beneficial to biologists studying the chronic effects of pollutants. Biologists believe that the mechanism which fish use in detoxifying toxaphene is the same used to neutralize the effects of other pollutants, but further studies are necessary before any firm conclusions can be drawn.

News release

VA Administrator Richard L. Roudebush reaffirmed the Veterans Administration's continued commitment to increased employment of handicapped persons, and has called for "added emphasis for additional progress in 1977."

"As a result of our concerted efforts to provide not just jobs, but careers for these deserving individuals, a total of 5.6 percent of VA's work force is now made up of handicapped persons," he said.

"The Rehabilitation Act of 1973 and the Vietnam Era Veterans Readjustment Act of 1974 mandate affirmative action programs for hiring the handicapped, and these have served both to augment and amplify VA's traditional leadership in this area," Roudebush said.

The VA Administrator pointed out that 6.6 percent of the agency's employees are disabled veterans. Disabled veterans and handicapped persons on the VA's rolls number more than one in ten of the 219,403 employees.

"While we at VA intend to do all we can, it is obvious that one agency cannot provide all the opportunities that are needed. I urge leaders in business and industry to join us in making jobs available to their handicapped fellow citizens," Roudebush said.

THE SWEETEST STORY EVER TOLD...

A Valentine

PHOTO

SEE PAGE 23
IN THIS WEEK'S NORTH KENT LEADER

NEED CALCIUM NITRATE?

OR UREA 45?

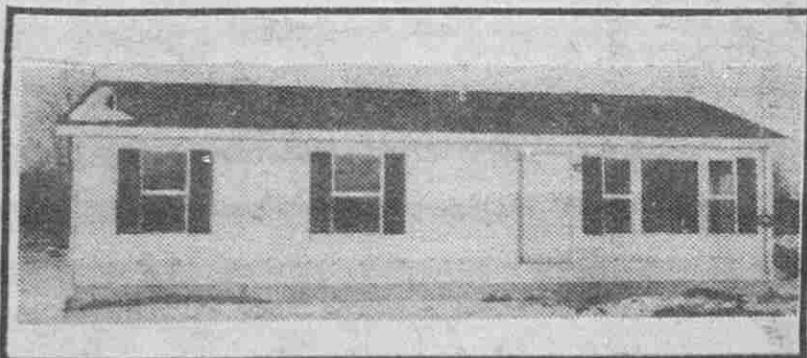
BULK OR BAG

Franklin Reister
& Sons

FOR MORE
INFORMATION CALL:
887-9689 or
887-9933

OPEN HOUSE

DIRECTIONS: 101-17 MILE RD.
1/2 MILE EAST OF RED PINE



NEW 3 BEDROOM RANCHER

1/2 ACRE \$23,900

Carriage
House Realty

Phone:
887-7307



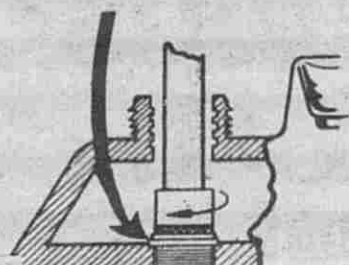
SPEED
LIMIT

55

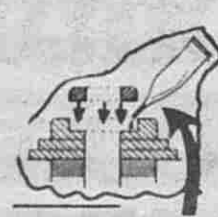
STOP DRIPPING FAUCETS

FOR LIFE*

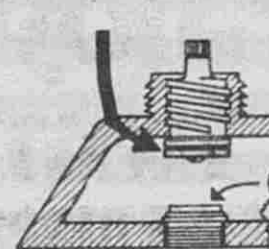
NO IF'S-ANDS-OR BUTS
HERE'S HOW WE DO IT FOR YOU



Our special tool
machines your
old seat



A Surgical
grade stainless
steel seat is in-
stalled with
space age ad-
hesive



A Delrin® wash-
er is fitted to
your shut off
stem.

Faucet drip is costing you MONEY. Did you know that a slowly dripping faucet can waste over 2000 gallons of water per year. If this is a hot water drip, there's additional waste in the energy used to heat this water. LITERALLY — money down the drain.

The real cause of faucet drip is a pitted and corroded valve seat; a normal condition in any conventional faucet after a few years of use. A washer replacement is a temporary fix at best, since the ragged valve seat quickly chews up the conventional rubber washer and the cycle starts all over again.

Our surgical grade stainless steel seat will not pit and corrode from normal chemicals and minerals present in all water supplies; so we eliminate washers damaged by corroded seats. Further, our unique DELRIN washer will not swell from heat and cut off your hot water supply, a common occurrence with conventional hard rubber washers. DELRIN is so tough, it is used to make gears.

TOTAL COST: \$5.00 PER SINGLE SHUTOFF - NO OTHER HIDDEN COSTS. WE'LL SCHEDULE OUR WORK AT YOUR CONVENIENCE.

Every job is regis-
tered and guaran-
teed for the life of
the present owner.
Any failure is our
total responsibility.

(616) 874-7330

Apt. owners & mgrs.
call for special
quantity rates.

Rockford, Mi.
(616) 874-7330
Sales Associates